ARtPUT: Autonomous Rover to Pick Up Trash

Progress Report 6: Finished CAD Design and Pushing Through Adversity

Bokre Samson, Dylan Shattan, Brendan Amoah-Daniels

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Materials and Methods

Materials:

- DFRobot Devastator Tank Mobile Robot Platform serving as the tracked mobile base
- Raspberry Pi 4 Model B as the primary onboard computer for vision, navigation, and logging
- Arduino UNO for low-level motor, servo, and sensor control
- USB RGB camera with depth-sensing capability for object detection and grasp planning
- Ultrasonic sensors for short-range obstacle detection
- NEO-6M GPS module for position logging and route tracking
- Digital compass (magnetometer) for heading correction
- Infrared sensor for bin fill-level detection
- Two-finger robotic gripper actuated by a servo motor
- 12V DC suction pump with relay module
- Suction-assisted collection bin with approximately 500–1000 mL capacity
- L298N motor drivers for drive motors
- Buck converters for voltage regulation
- Two 11.1 V Li-Po batteries to power onboard systems
- Wireless telemetry and onboard data logging storage
- Cones, measuring tape, stopwatch, and field notebook for controlled outdoor testing

Procedure:

1. Dataset & training: To train an object-detection model (YOLO/SSD-style), gather and label a significant number of photos of the target litter types in the target park. Use artificial occlusions to enhance.
2. Software stack: Put in place a perception pipeline (object detection on RGB-D + depth clustering), SLAM-based localization (Lidar + odometry), and a manipulation controller that translates detected object pose to a grasp technique. We'll test two modes of perception manipulation: Vision-only bounding-box grasping (A); vision plus depth-point-cloud grasp planning (B).
3. Three 10 m × 10 m test plots in a public park with low pedestrian traffic are chosen for the test. A fixed number (N = 20) of different litter items are placed in each plot at random locations (pre-approved with park authority). Mark the start and goal points.
4. Trials: For each mode (A, B) run 10 trials per plot (total 60 runs), each trial starting from the same location, time-limited to 15 minutes or until bin full. Rotate item arrangements between trials. Record video, sensor logs, and manual observer notes.
5. Baseline: Include 10 human-assisted pickup runs (human uses same route, picks items by hand) for performance comparison.

Data Collection Methods:

- Automated logs: timestamps of detections, attempted grasps, success/failure flags, battery consumption, distance travelled.
- Manual annotations: items missed, misclassifications, pedestrian interactions, environmental notes (wetness, wind).

- Post-run bin content check to count unique items collected.

Safety Concerns:

- The use of electronics and batteries in our project presents a risk of burns and electrical shock.
- The gears in our system may catch onto our limbs and clothing, potentially crushing our limbs.
- The rover may move erratically and hit someone, causing a harmful impact.
- We may drop heavy objects on our feet.

Risk Mitigation:

- We will always wear safety glasses, gloves, and proper footwear while working on our project.
- We will not wear baggy clothes that could potentially get caught in gears.
- There will always be a supervising adult while we are working on our project.
- The rover will always be fully powered off when working on it.

Data Analysis

Since the previous progress report, our team has continued refining both the mechanical design and computational systems of the ARTPUT rover. A major milestone achieved during this phase was the completion of a full CAD model of the robot using Fusion 360 (Figure 1). This model enabled us to systematically evaluate component placement, optimize spatial organization, and identify design inefficiencies. As a result, several adjustments were made to improve the feasibility of the build, particularly in terms of mounting configurations and overall structural layout.

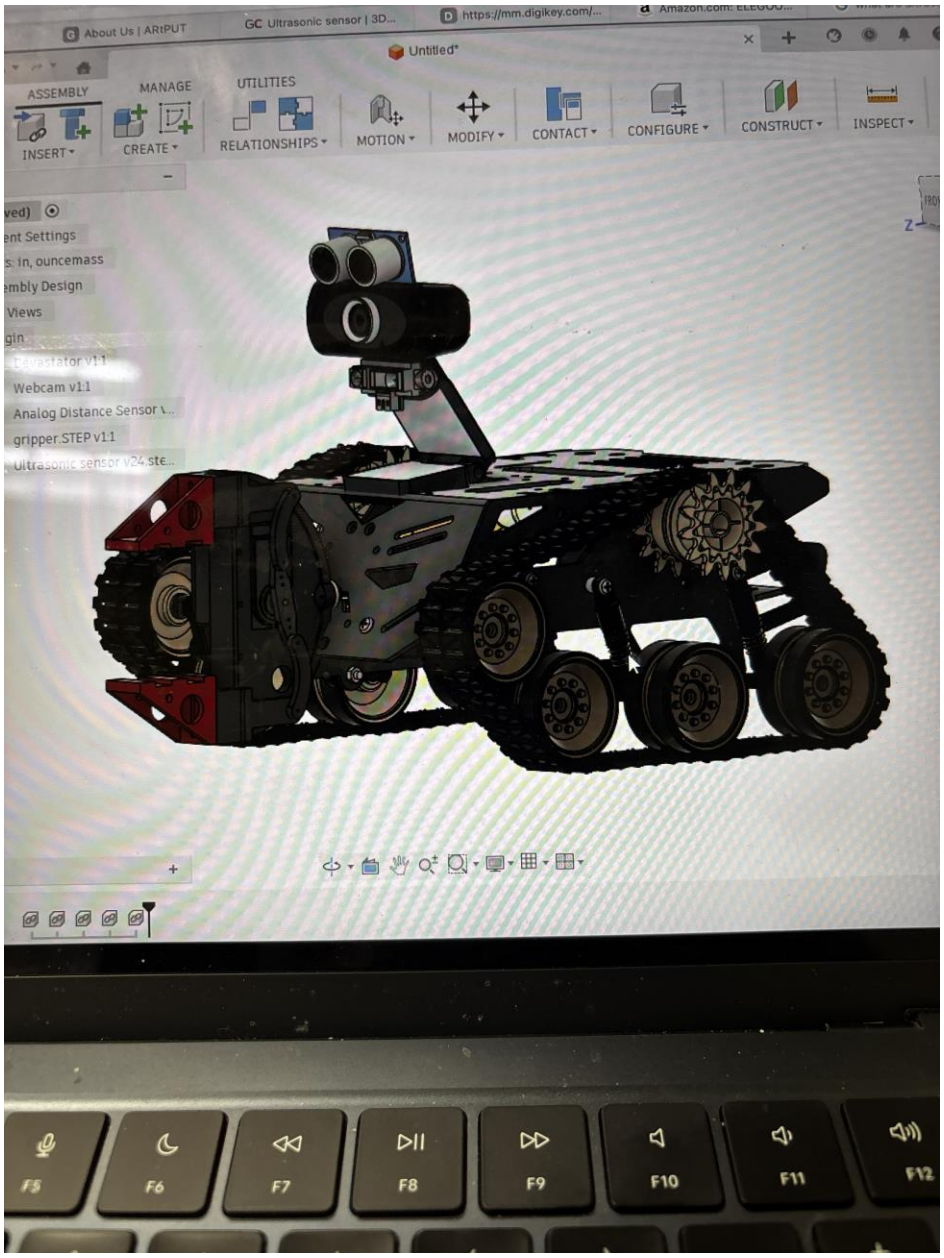

Figure 1. This is a picture of the CAD design of the ARTPUT rover showing the current model and what we expect the final robot to look like.

In the area of machine learning, we encountered a key limitation with our previous development platform, Roboflow. While Roboflow initially provided an accessible environment for training object-detection models, continued use of its core features required a paid

membership subscription. This created a constraint on our ability to iteratively train, test, and expand our model without incurring additional costs. Given the scope and timeline of our project, this dependency was not sustainable.

As a result, we made the decision to move away from Roboflow and look at different alternatives. Our current approach to solving this problem is to look at open-source models and frameworks available through GitHub. This gives us access to more algorithms without the need to subscribe to them, as well as more control over customization, training processes, and deployment. While more work goes into setting up the processes, we have more flexibility in terms of making our system more viable in the long run.

Meanwhile, we are working on ensuring that the physical build of the rover closely resembles the CAD design. This is done by translating the digital specifications of the rover into the physical world, ensuring that all the parts of the rover fit well with each other, and preparing for the integration of the other parts of the rover in the future. This is an important phase in bridging the gap between the design and the implementation of the rover.

Overall, this phase represents a strategic improvement in our technical approach and system design. By working to improve our shortcomings in our machine learning pipeline and continuing to improve our physical build, we are setting ourselves up well for integration and testing in the next phase.

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